

STABILITY



Stable - an object displaced will return to it's original position. (analogy - ball in cup)



Unstable - an object displaced will accelerate away from it's original position. (analogy - ball on hill)

Same in atm -

A layer in atm is stable when an ascending air parcel becomes cooler and/or more dense than surrounding environment. A stable atm will resist vertical motions. However, lifting can occur, but an external outside force is needed i.e., a wedge of cold air forcing another air mass to rise, or flow over a mountain. In other words, energy must be supplied to the parcel in order to lift it.

A layer in atm is unstable when an ascending parcel becomes warmer and/or less dense than the surrounding air and will accelerate upwards. Unstable air needs no energy input; it will rise on its own buoyancy! It will shoot up like a cork in water.

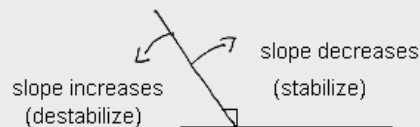
Good weather usually accompanies a stable atm, and cloudy, rainy weather occurs in unstable atm. But be careful!!! Stratus clouds can occur in stable atm, and sometimes light rain may fall. You may also have moist air overrunning a stable layer and have precip. Finally, sometimes upper level disturbances may trigger rain even if the surface layer is stable (more on this in a few lectures).

To figure out stability, need to measure vertical temp profile. This temp profile, called a sounding, varies from place to place and day to day. One method for determining stability is called the parcel method. Many other methods exist, including the popular stability indexes. We make two assumptions for the parcel method:

- 1) Assume a parcel is somehow displaced upwards (usually from the surface)
- 2) The parcel does not mix with surrounding air, and is therefore "insolated" from the environment. This means the parcel's motion will be adiabatic.

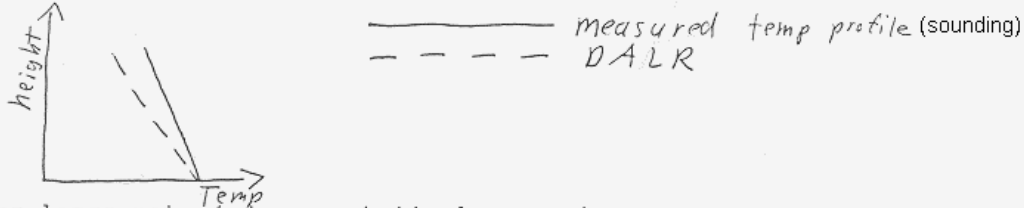
Recall definition for lapse rate: $-\frac{\Delta T}{\Delta Z}$

In meteorology, we define a sounding in which the temp decreases with height as having positive slope, and vice versa. We do this because we like to confuse math majors. Just kidding - the actual reason is because there is a negative sign "bogused" in the definition of a lapse rate, which switches the sign for a slope from the conventional math sense



Unsaturation motion

A displaced parcel will follow the dry adiabat. If the measured lapse rate is less than dry adiabat, the lifted parcel will have an internal temp less than the surrounding air, and will sink back to its original position.



measured lapse rate < dry adiabatic lapse rate

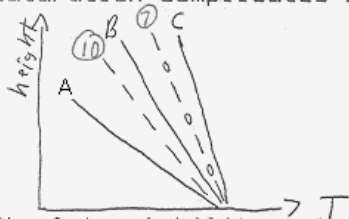
An unstable case is the reverse of the last argument.



measured lapse rate > dry adiabatic lapse rate.

Condensation affects stability

Saturation complicates stability.



A - Absolutely unstable

B - Conditionally stable

stable-unsaturated event
unstable-saturated event

C - Absolutely stable

SLR — 0 — 0 — 0 7 degrees C / km

DALR — — — — — 10 degrees C / km

Absolute stability exists when the measured lapse rate is less than the saturated (moist) lapse rate. It will be stable regardless of whether the atm is saturated or unsaturated. [Case C]

Absolute instability exists when the measured lapse rate is greater than the DALR. It will be unstable regardless of whether the atm is saturated or unsaturated. This type of instability cannot persist too long, because the atm will respond by overturning itself rapidly. This can occur in the desert, where there is strong surface heating. [Case A]

Conditional stability exists when the measured lapse rate is between the SLR and the DALR. If the atm is unsaturated, it is stable since it is less than the DALR. If the air becomes saturated (a cloud), it will be unstable since it has a greater lapse rate than the SLR. In other words, the release of latent heat makes the air parcel warmer than the surrounding air, and what was a stable situation is now an unstable situation, and the parcel has become buoyant. [Case B]

*****VERTICAL*MOISTURE**PROFILE*****

One must also consider how moisture content varies with height as well. The following simple example is associated with the development of severe storms. Consider this drawing:

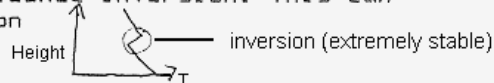


Note that the surface layer is very moist, and the top is unsaturated. If the layer is forced to rise, even a little, the upper portion of the layer cools at the DALR, while the bottom cools slower at the SLR. It does not take much lifting before the upper part of the layer is much colder than the bottom part. In other words, the sounding slope will increase with lifting in this case. The entire layer becomes absolutely unstable. This type of instability that results from unsaturated air over saturated air ^{being lifted} is called convective instability.

So what causes a stable atm?

1) Air warms aloft. Two ways:

a. This can be caused by subsidence, and since sinking air is compressed, it will warm and form a subsidence inversion. This can almost act as a lid!!! A typical inversion looks like this on a sounding.



An inversion is a layer in the atm is where temp actually increases with height (or negative slope). Physically, what happens is that a rising air parcel that encounters an inversion will find itself cooler than the surrounding environment and will sink back below the subsidence inversion. Some implications - subsidence inversions prevent pollutants from escaping into the air above, and in cities smog can form. Another subsidence inversion occurs at the tropopause, since generally air is sinking in the stratosphere. Remember earlier in the course I told you that the tropopause actually acts as a lid and ascending air spreads out below it - now you know why...

b. Air aloft might also be replaced by warmer air by wind transport.

2) Lower air cools. Can be caused by influx of cold air via wind transport, radiational cooling (earth radiating heat back to space at night), or air moving over a cold surface.

What causes an unstable atm?

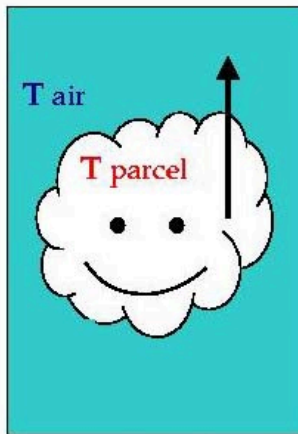
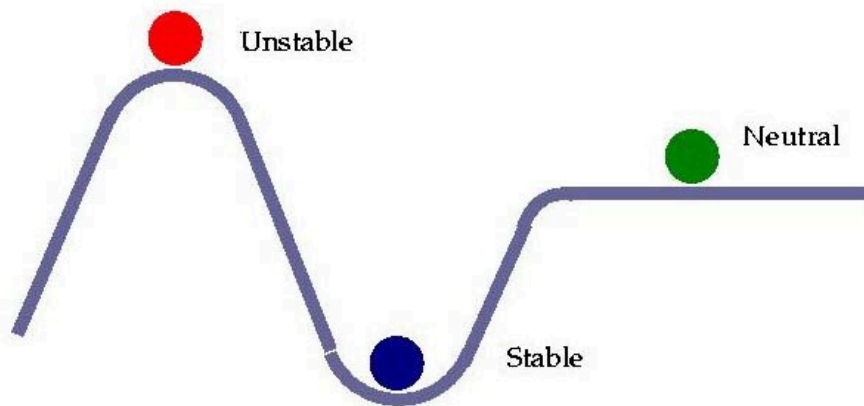
1) Air cools aloft

2) Air warms at surface. Some causes - daytime heating, influx of warm air, or moving over a surface warmer than itself.

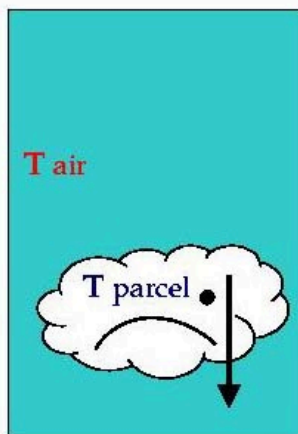
Of course, realize that vertical moisture distribution complicates stability (and instability) as well.

Final note - just because atm is unstable doesn't guarantee vertical accelerations. A "triggering mechanism" is needed. Recall the ball on a hill example: the ball needs to be disturbed somehow before instability takes over, or it will just sit there and do nothing.

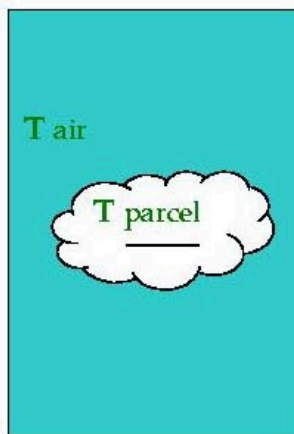
ATMOSPHERIC STABILITY



Unstable
 $T_{\text{parcel}} > T_{\text{air}}$
Parcel is lighter
and moves up.

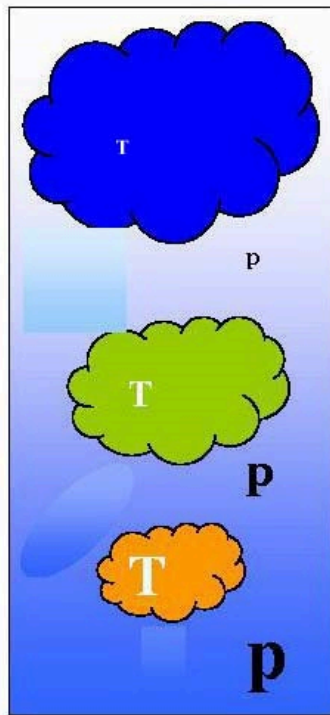


Stable
 $T_{\text{parcel}} < T_{\text{air}}$
Parcel is heavier
and moves down.



Neutral
 $T_{\text{parcel}} = T_{\text{air}}$
Parcel stays put.

ATMOSPHERIC STABILITY (continued)



Rising air parcel encounters decrease in surrounding pressure and expands.

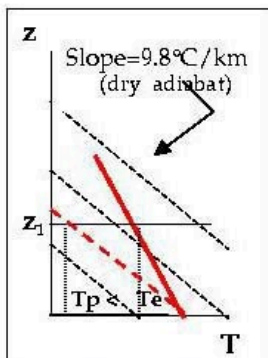
Expansion is adiabatic and implies a decrease in parcel temperature.

The dry adiabatic lapse rate:

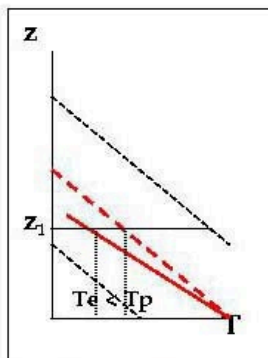
$$\Gamma_d = 9.8^\circ\text{C} / \text{kilometer}$$

Stability of Dry Air

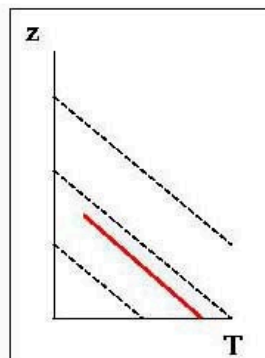
In the diagrams below solid red line is the environmental lapse rate, and dashed red is the dry adiabat for a parcel rising from the surface in that environment. T_p and T_e are the parcel and environment temperature at level Z_1 .



STABLE

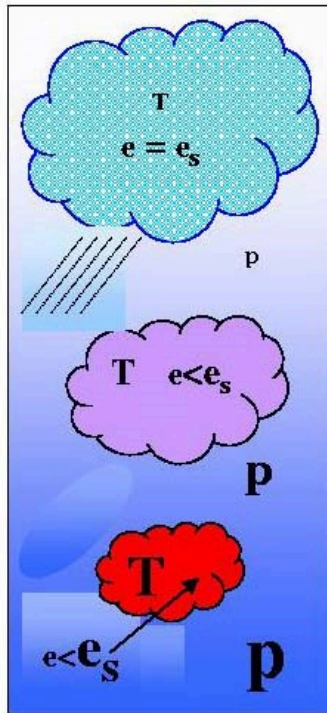


UNSTABLE



NEUTRAL

STABILITY OF MOIST AIR



Rising air parcel includes water vapor.

Adiabatic cooling reduces its ability to contain water.

Vapor condenses and releases latent heat.

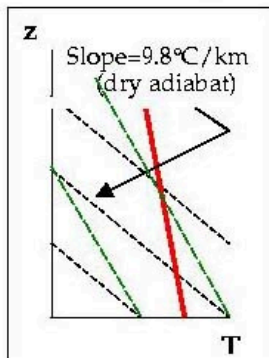
Cooling rate of air is reduced.

Moist adiabatic lapse rate:

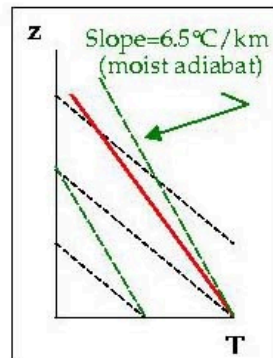
$$\Gamma_{\text{moist}} = 6.5^\circ\text{C} / \text{kilometer}$$

Stability of Saturated Air

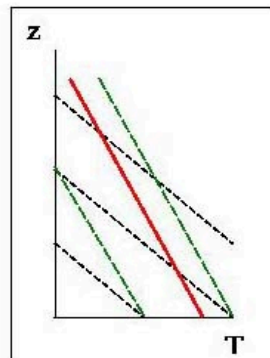
In the diagrams below solid red line is the environmental lapse rate.



STABLE



UNSTABLE



NEUTRAL

